SDS-001: Software Design Specification

Project:	Project Atlas (Master Flightline Clock)
Document ID:	SDS-001
Version:	1.0.0
Product Owner:	Peter Lochhead Hepburn
Date:	July 3, 2026

1. Introduction

This document provides a detailed technical specification for the "Master Flightline Clock" software, codenamed **Project Atlas**. It serves as the definitive reference for the system's architecture, timing logic, and persistence strategy, ensuring design integrity and GAMP-style traceability.

2. Architectural Overview

Atlas is implemented using a modular **Model-View-Controller (MVC)** pattern. The system is designed to run as a dedicated appliance on a Mac Mini M4, providing real-time timing control for competitive aviation events.

- **Core Engine:** Manages state and timing logic independent of UI.
- **Persistence Layer:** SQLite database for heat logs and hardware settings.
- **Operator Interface:** High-contrast, custom cockpit GUI for mission control.
- **Renderer:** Adaptive output layer for LED hardware or simulator.

3. Timing Engine & State Machine

The `AtlasEngine` operates on a 100ms primary ticker. It calculates the current phase and countdown by comparing the local system clock (UTC synced) against programmed heat timestamps.

3.1 Operational Phases

Phase	Name	Condition / Threshold	Display Meanings
0	ARMED	now < (track_open - 300s)	H# | Thermalling Direction
1	PRE-ENTRY	now > (track_open - 300s)	TRACK OPENS IN (Countdown)
2	TRACK ENTRY	now > track_open AND now < (track_open + 300s)	TIME TO ENTER (Countdown)
3	ACTIVE	now > (track_open + 300s) AND now < track_close	TRACK CLOSES IN (Countdown)
4	COOLDOWN	now > track_close AND now < heat_end	HEAT COMPLETED

4. Persistence Layer Strategy

Atlas utilizes an embedded SQLite database (`atlas.db`) to provide reliability and an automated audit trail.

4.1 Database Tables

- **heats:** Contains `heat_number`, `thermalling_dir`, `track_open`, `track_close`, `heat_end`, and `status` (SCHEDULED, ACTIVE, COMPLETED, CANCELLED).
- **settings:** Key-Value storage for `competition_name`, `display_width`, `display_height`, and `brightness`.

5. Adaptive Display Rendering

To maximize visibility on diverse LED matrix hardware (P5/P10 panels), the system implements an adaptive rendering layer.

- **Compact Mode (64x64):** Automatically triggered if width is less than 128. Abbreviates labels (e.g., "TRACK OPENS IN" becomes "OPENS") to maintain massive font sizes.
- **6-Row Geometry:** Content is mapped to specific vertical bands (Clock, Heat Info, Open Times, Primary Countdown, Status) to prevent data overlapping.
- **Intensity Control:** Supports live dimming (10% to 100%) mapped to environmental settings (Night, Overcast, Normal, Bright Day, Full Sun).

6. Error Handling & Safety

In alignment with GAMP 5 principles, Atlas prioritizes safety:

- **Graceful Shutdown:** Both operator and public displays are terminated simultaneously to prevent "stale" time displays.
- **Atomic Persistence:** Heat updates are written immediately to disk; a crash results in a "loss of 1s" maximum at restart.
- **Cancel Logic:** Active heats are flagged as `CANCELLED` in the logs rather than deleted, preserving event history.